Lab 3 Report

Compensator Design

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Contents

1 Introduction

The objective of this lab is to enhance the controller developed in Lab 2 by designing and testing a first-order compensator that meets the required performance specifications and improves the overall robustness of the KittiCopter system.

1.1 Background

In Lab 2, the KittiCopter was modeled as a simplified rotorcraft influenced by aerodynamic drag, gravitational forces, and rotor thrust. A free-body diagram was used to represent these forces, enabling the derivation of a second-order differential equation describing the vertical motion of the system.

Since the mass of the KittiCopter was normalized to unity, the governing equation simplifies to:

$$\ddot{y}(t) = f_t(t) - b\dot{y}(t) - g$$

where $\ddot{y}(t)$ is the vertical acceleration, $f_t(t)$ is the rotor thrust, $b\dot{y}(t)$ represents the aerodynamic drag (proportional to velocity), and g is the constant gravitational acceleration. This equation forms the foundation for modeling the system dynamics.

1.2 System Modelling

By applying the Laplace transformto the linearized model, the transfer function relating the input thrust to the vertical position output was obtained. The system was approximated as a first-order model, dominated by a single time constant, which simplifies analysis and controller design.

To determine the model parameters experimentally, a step response test was performed. A known step input was applied, and the output position was measured. From this response, the *time constant* (τ), *system gain* (K), and *sensor gain* (K_s) were calculated.

The results obtained are summarized below:

$$\tau = 10.02 \text{ s}, \quad K = 21.6292, \quad K_s = 0.96 \text{ V/m}$$

Using these values, the first-order transfer function of the KittiCopter system is expressed as:

$$G(s) = \frac{21.6292}{10.02s + 1}$$

This transfer function accurately represents the system dynamics and serves as the foundation for designing the compensator in this lab. A block diagram of the complete KittiCopter control system is shown in Figure 1.

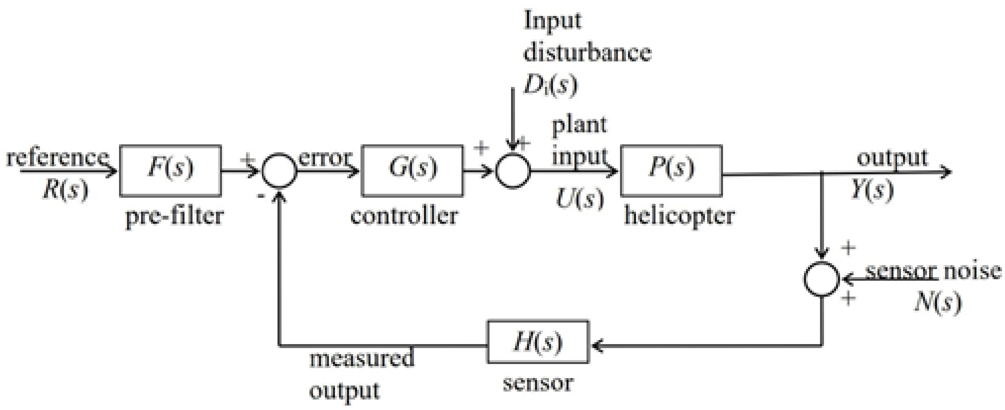

Figure 1: Block diagram of the KittiCopter system showing the various signals.

2 Compensator Design

The objective of this section is to design a first-order compensator to improve the Kitticopter system's transient response, stability, and tracking performance, building upon the controller developed in Lab 2. This compensator ensures accurate set-point tracking, reduced overshoot, faster settling, and robust disturbance rejection.

2.1 Performance Specifications

The compensator must satisfy the following requirements:

- **Set-point tracking:** within 0.4 m of the final value.
- **Overshoot:** less than 20% (targeted 10% for design margin).
- **Settling time:** within 2% of final value, $t_s < t_e$. Although the experimental settling time was 8 s, a faster settling time of 6 s was used for calculations.
- **Disturbance rejection:** output should return to within 2% of the initial value in less than t_e.
- **Sinusoidal tracking:** track a 0.1 rad/s sinusoidal input within 0.4 m.

2.2 Controller Selection

In Lab 2, a proportional controller was implemented to stabilize the Kitticopter. However, several limitations were observed:

- Tracking accuracy did not exceed 90%, below the performance requirement.
- Lack of robustness, with poor disturbance rejection.
- Slow system response.
- Sensitivity of the potentiometer made precise input adjustment difficult.

To overcome these issues, a lead compensator was chosen. A lead compensator improves the system's transient response and stability by introducing phase lead in the frequency response, increasing the system's bandwidth and performance.

The transfer function of a lead compensator is given by:

$$C(s) = K\frac{s+z}{s+p}$$

where:

- K is the compensator gain,
- z is the zero of the compensator,
- p is the pole of the compensator ($z < p$ for phase lead).

Key advantages of a lead compensator include:

- **Reduced Overshoot:** improves damping ratio to meet performance criteria.
- **Faster Response:** shifts system poles to the left, reducing settling time.
- **Improved Phase Margin:** enhances system stability and reduces oscillations.
- **Increased Bandwidth:** allows faster response to higher-frequency inputs.
- **Improved Disturbance Rejection:** maintains accuracy even under external perturbations.

Thus, the lead compensator addresses the limitations of the proportional controller and improves the Kitticopter's overall control performance. While the compensator is designed to meet the performance specifications outlined in the previous subsection.

2.3 Design Calculations

2.3.1 Pole Selection for Settling Time

For a first-order system, the settling time (t_s) to within 2% of the final value is approximated by:

$$t_s \approx 4\tau_{\text{eff}}$$

The experimentally observed settling time for the Kitticopter system was 8 s. However, to ensure the designed controller does not operate on the borderline and to account for tolerances and uncertainties, a slightly faster design settling time of 6 s was used for calculations:

$$\tau_{\text{eff}} = \frac{t_s}{4} = \frac{6}{4} = 1.5\,\text{s}$$

The corresponding real pole location for the chosen settling time is:

$$P_{\text{real}} = -\frac{1}{\tau_{\text{eff}}} = -0.667 \text{ rad/s}$$

2.3.2 Overshoot and Damping

To ensure robustness and account for tolerance issues experienced in Lab 2, a conservative target maximum overshoot of $OS = 10\%$ was chosen, instead of the standard 20%. This results in a damping ratio:

$$OS = 100\,e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \quad \Rightarrow \quad \zeta = 0.591$$

The corresponding damping angle is:

$$\theta = \arccos(\zeta) = 53.8°$$

The imaginary part of the desired closed-loop pole is calculated as:

$$\omega_d = \tan(\theta) \cdot |P_{\text{real}}| = \tan(53.8°) \cdot 0.6667 = 0.91093$$

Hence, the desired complex pole locations for the lead compensator design are:

$$s = -0.6667 \pm j0.91093$$

This choice of reduced overshoot ensures that the system remains well within acceptable performance margins, improving stability and robustness.

2.3.3 Lead Compensator Design

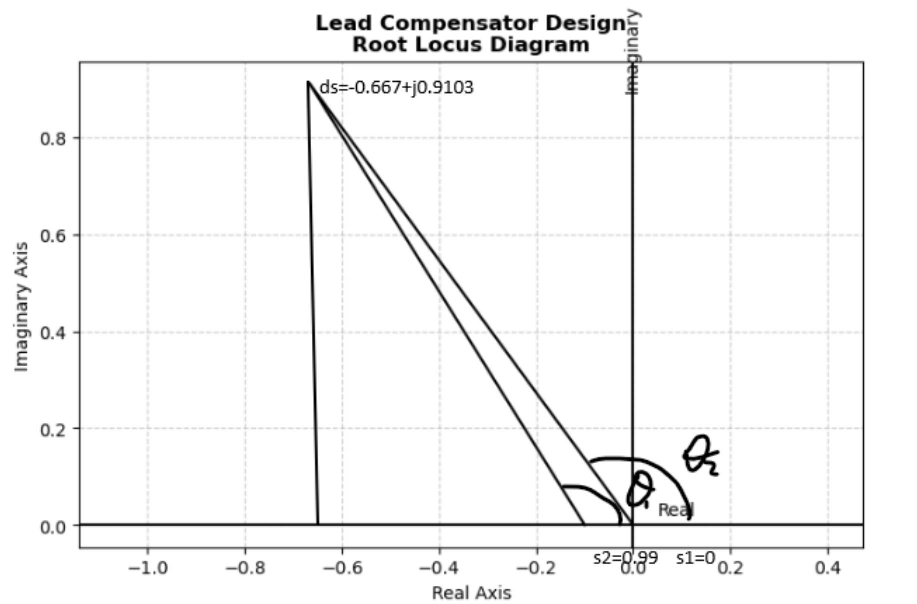

Figure 2: Diagram of Root Locus compensator design

The root locus of the uncompensated system does not pass through the desired pole location, so a lead compensator is introduced with transfer function:

$$C(s) = K\frac{s+z}{s+p}$$

Using the calculated values for the design:

$$\omega_d = 0.91093, \quad P_{\text{real}} = 0.6667, \quad P_{\text{offset}} = 0.0998$$

The phase angles are computed as:

$$\theta_1 = 180 - \arctan\left(\frac{0.91093}{0.6667 - 0.0998}\right) = 121.89^\circ$$

$$\theta_2 = 180 - \arctan\left(\frac{0.91093}{0.6667}\right) = 126.2^\circ$$

The total angle from existing poles and zeros:

$$\sum \theta_z - \sum \theta_p = 0 - 121.4 - 126.2 = -248.095^\circ$$

For the desired closed-loop pole to lie on the root locus, the compensator must contribute:

$$\theta_z - \theta_p = -180 - (-247.6) = 68.0953^\circ$$

Minimum zero location:

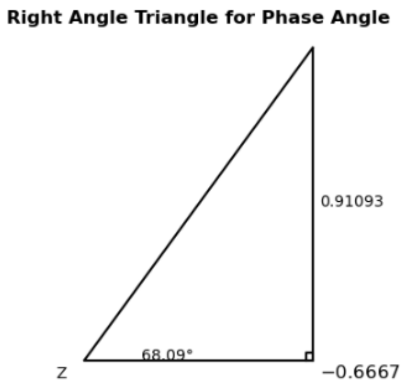

Figure 3: Diagram for minimum zero location

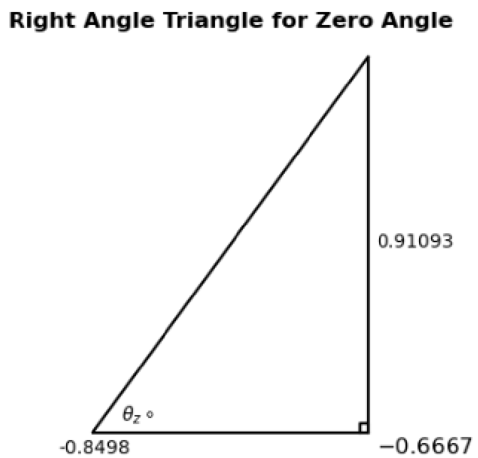

Figure 4: Diagram for zero angle contribution

The zero is positioned so that its contribution provides the full required 68.1° phase. Using trigonometry:

$$Z = -0.6667 - \frac{0.91093}{\tan(68.1^\circ)} = -1.03297$$

A safe zero location is chosen within the range $-1.03297 < z < -0.6667$:

$$z = -0.849835$$

The zero angle contribution:

$$\theta_z = \arctan\left(\frac{0.91093}{0.849835 - 0.6667}\right) = 78.63^\circ$$

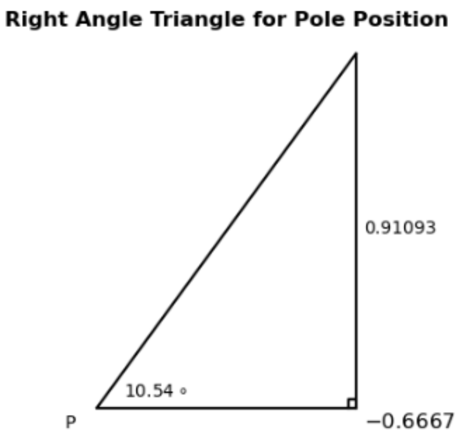

Figure 5: Diagram for pole position

The pole angle contribution:

$$\theta_p = \theta_z - 68.1 = 10.5373°$$

The corresponding pole position:

$$p = -0.6667 - \frac{0.91093}{\tan(10.53°)} = -5.5623$$

Thus, the lead compensator is defined as:

$$C(s) = K\frac{\frac{1}{0.8498}s+1}{\frac{1}{5.56}s+1} = K\frac{1.1767s+1}{0.1799s+1}$$

The compensator gain at the desired pole is calculated by substituting the desired complex pole $s = -0.6667 + j0.91093$ into the formula:

$$K = \left|\frac{1}{G(s)C(s)}\right|$$

Substituting the values:

$$G(s) = \frac{21.6292}{10.02s+1}, \quad C(s) = \frac{1.1767s+1}{0.1799s+1}, \quad s = -0.6667 + j0.91093$$

$$K = \left|\frac{1}{\frac{21.6292}{10.02(-0.6667+j0.91093)+1} \cdot \frac{1.1767(-0.6667+j0.91093)+1}{0.1799(-0.6667+j0.91093)+1}}\right| \approx 0.602$$

For simplicity in implementation, the gain is set to:

$$K = 1$$

2.4 Controller Test in Simulation

A step response simulation was performed using the Simulink model shown in Figure 6. The system was tested with a step input of amplitude 4.8 (5 x 0.96), and the results were evaluated in terms of accuracy, overshoot, settling time, and disturbance rejection. The resulting step response is presented in Figure 7.

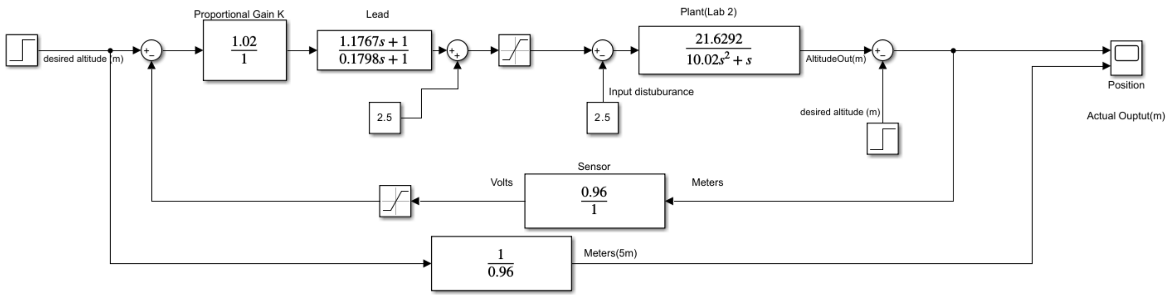

Figure 6: Simulink model for the system with compensator and plant

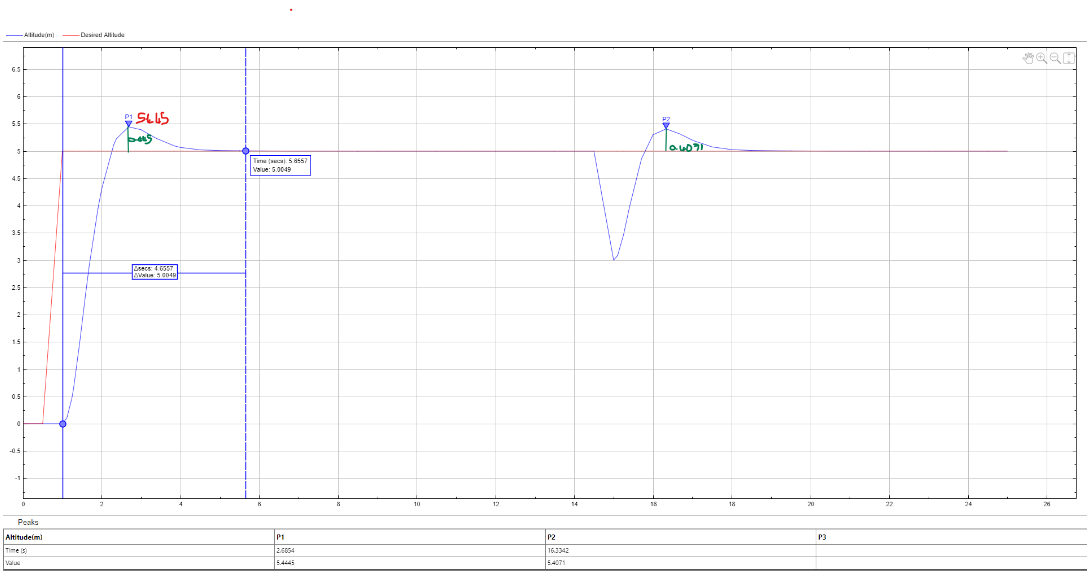

Figure 7: Simulated step response of the system.

The accuracy of the system was first evaluated by examining the final value of the step response. As observed in Figure 7, the system tracks the step input with a final steady-state value of

$$y_{\text{final}} = 5.0049, \tag{1}$$

which corresponds to a tracking accuracy of

$$\text{Percentage error} = \frac{5.0049 - 5}{5} \times 100 = 0.098\%. \tag{2}$$

This comfortably exceeds the required accuracy of 92%.

Next, the overshoot was measured. The maximum value reached by the step response is

$$y_{\max} = 5.445, \tag{3}$$

which results in an overshoot of

$$\%OS = \frac{y_{\max} - 5}{5} \times 100 = 8.9\%. \tag{4}$$

This overshoot is well below the specified maximum limit of 20%, indicating that the system is well-damped.

The settling time, defined as the time required for the response to remain within ±2% of the final value, was determined to be

$$T_s = 4.6557 \text{ s}, \tag{5}$$

which is significantly less than the specified maximum of 8 s, demonstrating a fast and stable response.

To evaluate disturbance rejection, an output disturbance of magnitude 1 was introduced at $t = 15$ s. The system output initially peaked at

$$y_{\text{disturbance, peak}} = 5.4071, \tag{6}$$

before returning to its steady-state value of 5.0049 within a short period. This indicates effective disturbance rejection and strong compensator robustness, as the system recovered well within the allowable settling time.The tracking error for the sinusoidal input is less than 0.4 m, which satisfies the required specification. The compensator thus demonstrates reliable performance in tracking both step and sinusoidal inputs while maintaining minimal error.

Overall, the compensator exhibits excellent dynamic performance and robustness, making it well-suited for the control objectives of the system.

2.5 Physical Design

The schematic of the circuit with these values is shown in Figure 8.

The lead compensator was realized using operational amplifiers and RC components, following the standard form:

$$C(s) = \frac{R_4}{R_3} \cdot \frac{R_1 C_1 s + 1}{R_2 C_2 s + 1}$$

From the design calculations, the compensator transfer function is:

$$C(s) = 1\frac{1.1767s + 1}{0.1799s + 1}$$

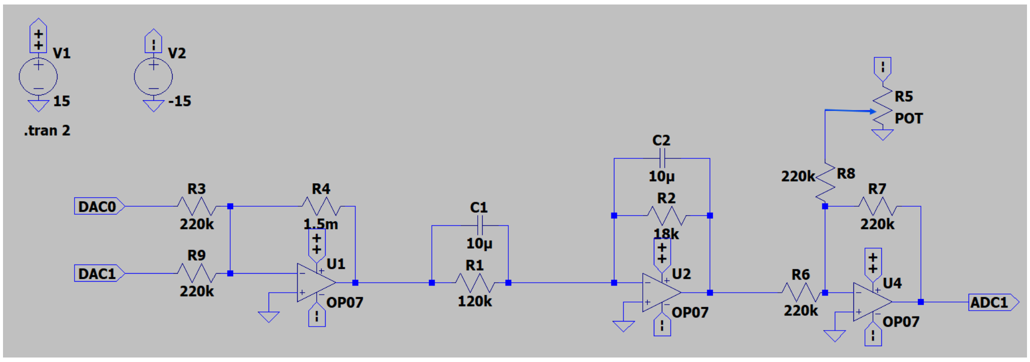

Figure 8: Lead compensator circuit diagram integrated with the controller circuit from lab 2

Step 1: Determine RC products Matching the transfer function to the circuit form:

$$R_1C_1 = \frac{1}{1.1767} = 0.8498\,\mathrm{s}, \quad R_2C_2 = \frac{1}{0.1799} = 5.56\,\mathrm{s}$$

Step 2: Select capacitor values Choose standard capacitor values:

$$C_1 = C_2 = 10\,\mu\mathrm{F}$$

Step 3: Solve for resistor values

$$R_1 = \frac{zero}{C_1} = \frac{1.1767}{10 \times 10^{-6}} \approx 117.6\,\mathrm{k}\Omega$$

$$R_2 = \frac{pole}{C_2} = \frac{0.1798}{10 \times 10^{-6}} \approx 17.9\,\mathrm{k}\Omega$$

Choosing practical E12 series values:

$$R_1 = 120\,\mathrm{k}\Omega, \quad R_2 = 18\,\mathrm{k}\Omega$$

Step 4: Determine overall gain resistors The overall circuit gain is:

$$\frac{R_2}{R_1} = 0.1528 \quad \Rightarrow \frac{R_4}{R_3} = 6.54$$

However, due to the previous proportional control design, choose $R_3 = 220\,\mathrm{k}\Omega$. Then:

$$R_4 = (220k) \cdot (6.54) = 1.438\mathrm{M}\Omega$$

Choosing practical E12 series values:

$$R_4 = 1.5\,\mathrm{M}\Omega$$

Given $R_1 = 120\,\mathrm{k}\Omega$, $R_2 = 18\,\mathrm{k}\Omega$, $R_3 = 220\,\mathrm{k}\Omega$, and $R_4 = 1.5\,\mathrm{M}\Omega$, the overall gain is

$$G_{\text{overall}} = \frac{R_4}{R_3} \times \frac{R_2}{R_1} = \frac{1.5 \times 10^6}{220 \times 10^3} \times \frac{18 \times 10^3}{120 \times 10^3} = (6.818) \times (0.15) = 1.0227.$$

$$\boxed{G_{\text{overall}} \approx 1.02}$$

Thus, a gain of 1.02 was used in the simulink model of the circuit as seen in Figure 6.

After testing in simulation the components were soldered onto the veroboard as seen in Figure 11.

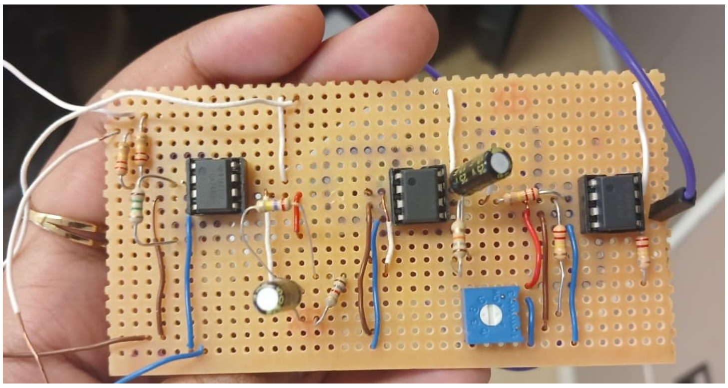

Figure 9: Final design on veroboard

3 Controller Testing

3.1 Tests Performed on the Controller

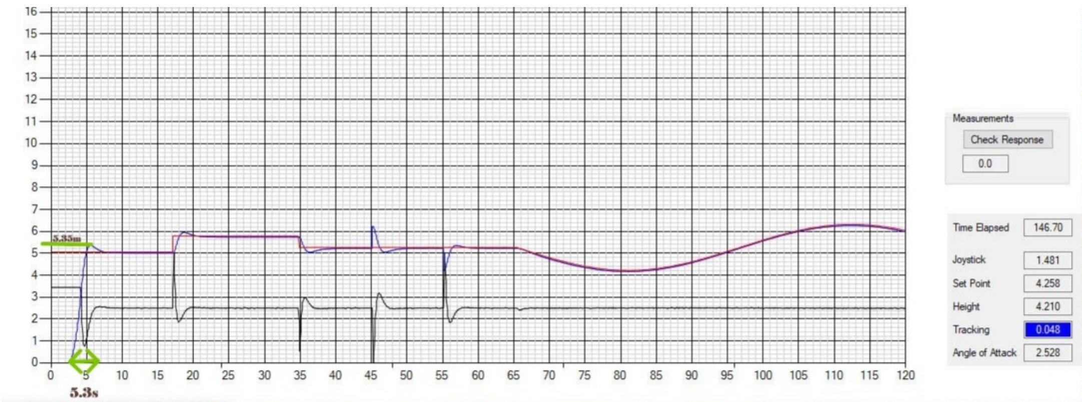

Figure 10: Experimental results of the step response test and sinusodial test.

The compensator underwent several tests to verify that it met the required performance specifications. A step input of amplitude 5 was initially applied to evaluate the system against the following criteria: accuracy greater than 92%, overshoot less than

20%, and settling time within 2% of the final value in under 8 seconds. Additionally, an output disturbance of amplitude 1 was introduced to examine whether the system could reject disturbances within the specified settling time. The results of the step response test are shown in Figure 10.

As shown in Figure 10, the compensator was subjected to multiple disturbances and step inputs of varying magnitudes to assess its robustness under diverse conditions. A sinusoidal input with amplitude 1 and frequency 1 rad/s was also applied to determine whether the system could track the input with high accuracy.

3.2 Performance Evaluation

The performance of the compensator was evaluated using the experimental results and compared against the design specifications. The steady-state value of the step response was measured as

$$y_{\text{final}} = 5.005, \tag{7}$$

which corresponds to a tracking error of

$$\text{Tracking error} = 5.005 - 5 = 0.005 \quad \text{(or 0.1\%)}, \tag{8}$$

well below the required threshold of 0.4 m, confirming accurate setpoint tracking.

The maximum response observed was

$$y_{\text{max}} = 5.35, \tag{9}$$

resulting in an overshoot of

$$\%OS = \frac{5.35 - 5}{5} \times 100 = 7\%. \tag{10}$$

This is comfortably within the specified maximum of 20%, indicating a well-damped response.

The system reached within 2% of the final value at $t = 5.3$ s, yielding a settling time of

$$T_s = 5.3 \text{ s}, \tag{11}$$

which is significantly below the maximum allowable value of 8 s.

An output disturbance was introduced at $t = 55$ s, causing the response to peak at

$$y_{\text{disturbance, peak}} = 5.2, \tag{12}$$

before returning to steady state at $y_{\text{final}} = 5.005$. The disturbance rejection settling time was calculated as

$$t_\delta = 58 \text{ s} - 55 \text{ s} = 3 \text{ s}, \tag{13}$$

well within the specified 8 s limit, confirming effective disturbance rejection.

For the sinusoidal input, the tracking error was measured at 0.048 m, which is below the 0.4 m requirement, demonstrating reliable periodic signal tracking.

Overall, the experimental results confirm that the lead compensator meets all design specifications in terms of accuracy, overshoot, settling time, disturbance rejection, and sinusoidal tracking. Minor deviations from the MATLAB simulation were observed, highlighting the importance of careful component selection and design considerations to maintain robustness in practical implementation.

3.2.1 Comparison of Experimental and Simulated Results

Table 1: Comparison between experimental and simulated controller performance

Performance Metric	Experimental	Simulated
Steady-state value	5.005	5.0049
Tracking error	0.005 (0.1%)	0.0049 (0.098%)
Maximum overshoot	5.35 m (7%)	5.445 m (8.9%)
Settling time	5.3 s	4.6557 s
Disturbance peak	5.2 m	5.4071 m
Disturbance settling time	3 s	3.5 s
Sinusoidal tracking error	0.048 m	< 0.4 m

The table highlights that the experimental results closely match the simulation, with minor differences attributable to real-world component tolerances and physical damping effects. Overall, both results confirm that the lead compensator performs reliably and meets all specified performance requirements.

3.3 Recommendations for Future Iterations

To enhance future iterations of the experiment, optimizing the lead compensator design could significantly improve the system's performance. The current compensator has a zero at 0.8498s and a pole at 5.56s. Fine-tuning the placement of these poles and zeros can further refine the transient response, reducing overshoot and settling time to improve overall system performance.

Additional damping through derivative control and careful zero placement could further minimize overshoot without significantly affecting response time. Using high-precision operational amplifiers would reduce non-ideal component effects, decrease noise, and improve system stability across varying operating conditions.

Advanced filtering techniques, such as Kalman filters or state estimators, could enhance disturbance rejection and speed up system stabilization. Finally, extending sinusoidal input tests to higher frequencies would allow evaluation of the compensator's bandwidth and tracking capability, ensuring reliable performance in more dynamic scenarios and real-world applications with rapid input changes.

4 Discussion and Conclusion

The objective of this lab was to design and implement a lead compensator to improve the performance of the Kitticopter control system and meet the following performance specifications:

- Tracking accuracy within 0.4 m of the final value.
- Maximum overshoot less than 20%.
- Settling time within 2% of the final value in less than 11 seconds.
- Disturbance rejection settling time under 11 seconds.
- Sinusoidal input tracking accuracy within 0.4 m for a 0.1 rad/s sinusoid.

The system was evaluated through both simulation and experimental testing. The experimental results showed a tracking error of 0.048 m, well below the 0.4 m threshold, confirming that the setpoint tracking requirement was met. The maximum overshoot observed was 5.35 m, which corresponds to approximately 7% overshoot—comfortably within the specified limit. The settling time was measured at 5.3 seconds, significantly below the 8-second requirement. Disturbance rejection was achieved in 3 seconds, well within the allowable limit, and sinusoidal input tracking demonstrated an error of 0.048 m, satisfying the required specification.

While the compensator successfully met all performance requirements, the experimental results showed minor deviations from the simulated response, which was slightly more robust. This highlights the importance of placing closed-loop poles sufficiently far from system boundaries during the design phase to maintain stability and robustness despite component tolerances and modeling inaccuracies.

For future improvements, the pole-zero placement could be further optimized. Adjusting the positions of the pole and zero or implementing a cascaded lead compensator could enhance phase margins, reduce overshoot, and improve transient response. The use of higher-precision operational amplifiers would help minimize non-ideal effects, reduce noise, and improve overall system stability.

In conclusion, the lead compensator effectively improved the Kitticopter's performance, meeting all specified requirements. Future design refinements and careful hardware selection could further enhance robustness and ensure improved handling of disturbances and dynamic inputs.

A Appendix and Notes

A.1 Matlab Code

```
% Define the numerator and denominator of the plant
numeratorp = 21.6292;
denominatorp = [10.02 1];  % 10.02s + 1

% Define the numerator and denominator of the compensator
numeratorc = [1.1767 1];   % 1.1767s + 1
denominatorc = [0.1799 1]; % 0.1799s + 1

% Create the transfer functions for the plant and compensator
plant = tf(numeratorp, denominatorp);
Comp = tf(numeratorc, denominatorc);

% Multiply plant and compensator to get the open-loop transfer function
sys = series(plant, Comp);  % Alternatively, sys = plant * Comp

% Define the sensor as a constant transfer function
sensor = tf(0.96, 1);  % H(s) = 0.96

% Create the closed-loop system with feedback
closed_loop_sys = feedback(sys, sensor);  % sys in forward path,
    sensor in feedback

% Plot the root locus of the open-loop system
figure;
rlocus(sys);
title('Root Locus of the Open-Loop System');
grid on;

% Choose two specific gains
gain1 = 0.63;
gain2 = 1;

% Calculate the poles for the closed-loop system at the specified gains
poles_at_gain1 = rlocus(sys, gain1);
poles_at_gain2 = rlocus(sys, gain2);

% Annotate the poles for gain = 0.63
hold on;
plot(real(poles_at_gain1), imag(poles_at_gain1), 'rx', 'MarkerSize',
    10, 'LineWidth', 2);
for i = 1:length(poles_at_gain1)
    text(real(poles_at_gain1(i)) + 0.1, imag(poles_at_gain1(i)), ...
        ['Gain 0.63: ', num2str(poles_at_gain1(i))], 'FontSize', 12,
            'Color', 'red');
end

% Annotate the poles for gain = 1
plot(real(poles_at_gain2), imag(poles_at_gain2), 'bx', 'MarkerSize',
    10, 'LineWidth', 2);
for i = 1:length(poles_at_gain2)
```

```
    text(real(poles_at_gain2(i)) + 0.1, imag(poles_at_gain2(i)), ...
        ['Gain 1: ', num2str(poles_at_gain2(i))], 'FontSize', 12,
            'Color', 'blue');
end
hold off;

% Display the corresponding poles for both gains
disp(['Poles for the closed-loop system at gain = ', num2str(gain1),
    ':']);
disp(poles_at_gain1);
disp(['Poles for the closed-loop system at gain = ', num2str(gain2),
    ':']);
disp(poles_at_gain2);
```

A.2 Additional Figures

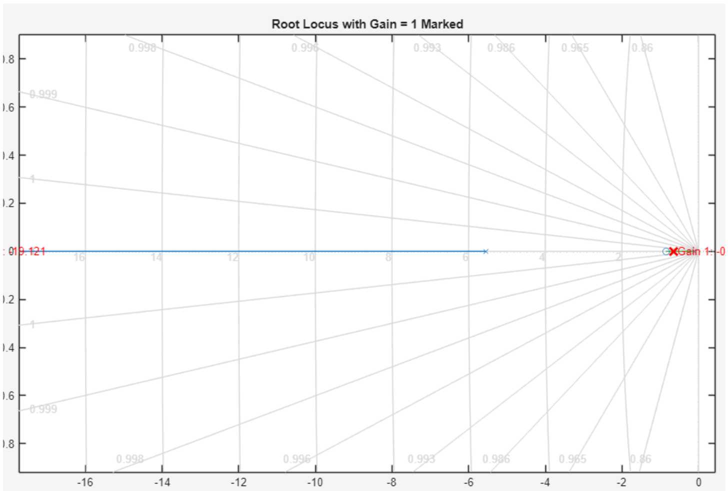

Figure 11: Root locus with a gian of 1

A.3 References

1. Course Notes, EEE3094S.
2. Lab 2 Report, MNLIMT001.
3. Lab 3 Report, EEE3094S.
4. Lab 3 Circuit Appendix, EEE3094S.